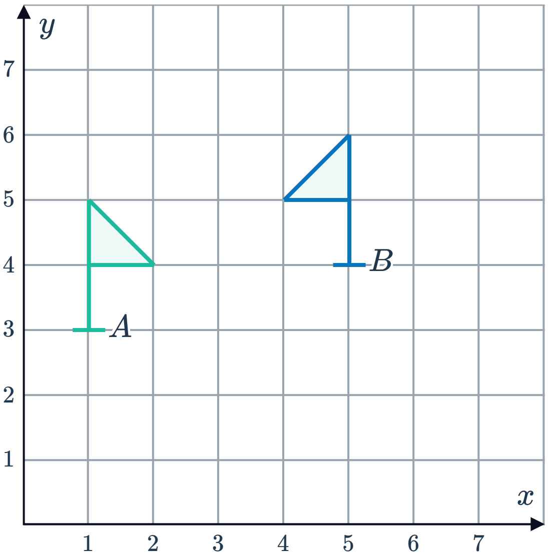
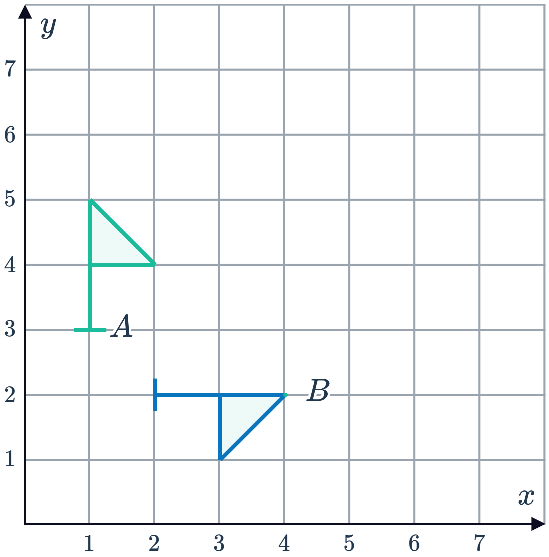
Composing transformations

1 For each of the following, state the two types of transformations that would be needed to get from Flag A to Flag B:

a



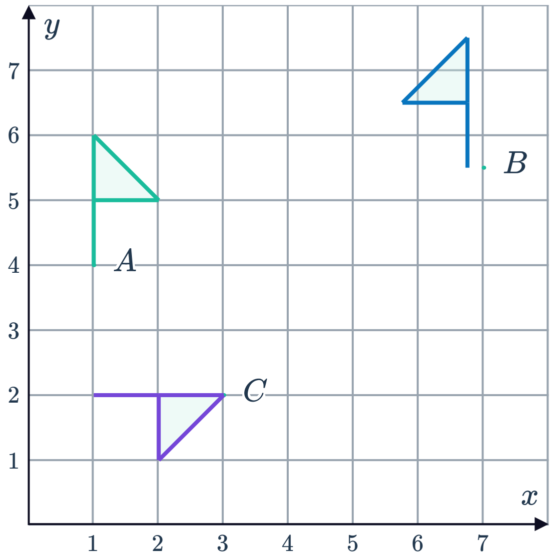
b



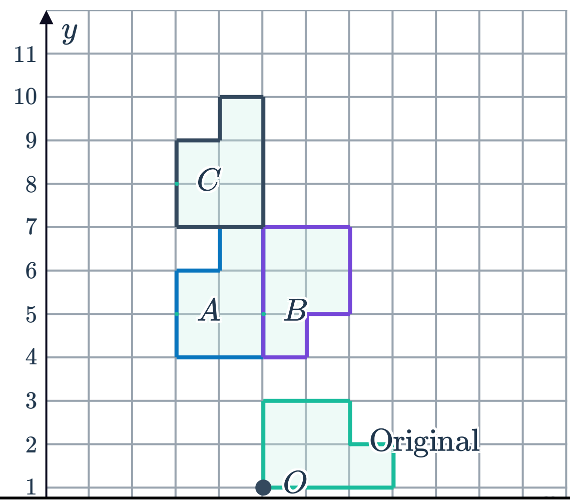
2 State the two types of transformations that would be needed to get from:

a Flag A to Flag B.

b Flag A to Flag C.

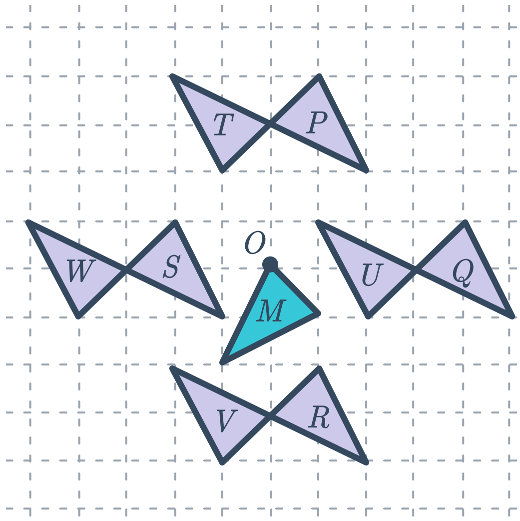


3 When the original shape is rotated 90° anti-clockwise about point O and then translated 3 units up, which of the shapes shown is the new shape?

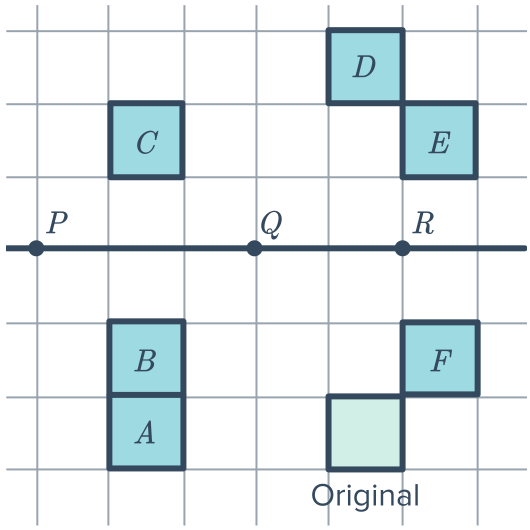


4 Triangle M is to be rotated 90° clockwise about point O, then translated 3 units left.

Which triangle corresponds to the result of this transformation?

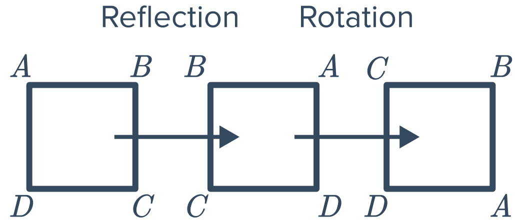


5 Identify the correct square when the original is rotated 90° anticlockwise about point Q, then reflected across the line PR, and then translated 4 units to the left.



6 Square $ABCD$ is reflected across a vertical line and then rotated 90° clockwise about its centre. This process is repeated with each resulting square. The result of the first set of reflection and rotation is shown:

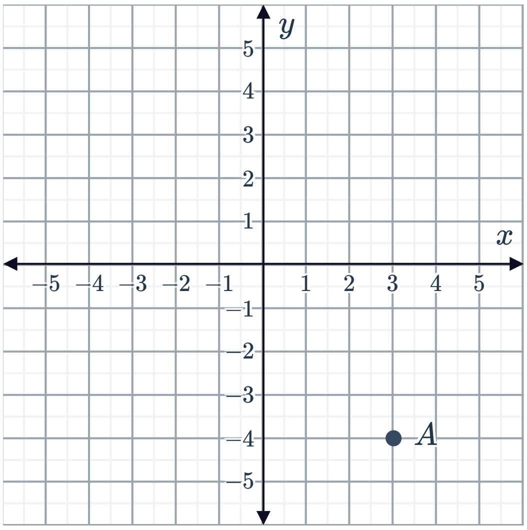
After 13 repetitions of this process, write down the vertices, clockwise from the top left corner.



7 Point A on the plane is to undergo two successive transformations:

a Plot point A', which is the result when point A is translated 4 units left.

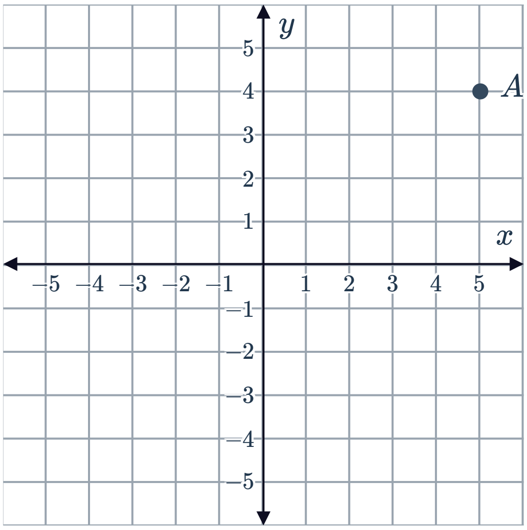
b Plot point A'', which is the result when point A' is rotated 90° clockwise about the origin.



8 Point A on the plane is to undergo two successive transformations:

a Plot point A', which is the result when point A is translated 5 units downwards.

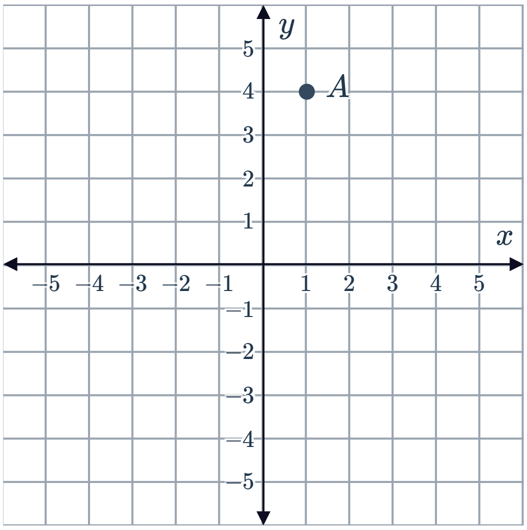
b Plot point A'', which is the result when point A' is rotated 90° anticlockwise about the origin.



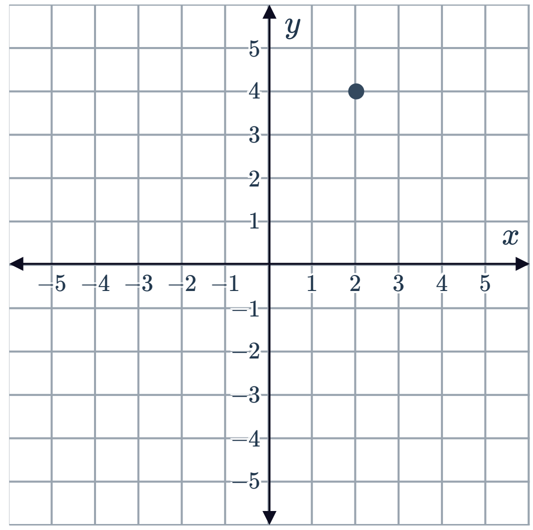
9 Point A on the plane is to undergo two successive transformations:

a Plot point A', which is the result when point A is translated 3 units right and 5 units down.

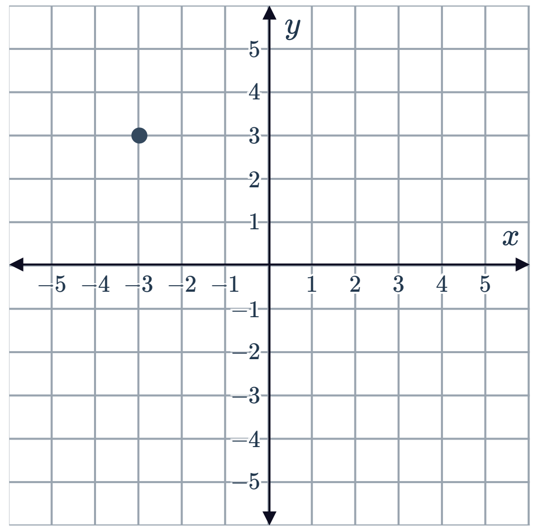
b Plot point A'', which is the result when point A' is rotated 90° anticlockwise about the origin.



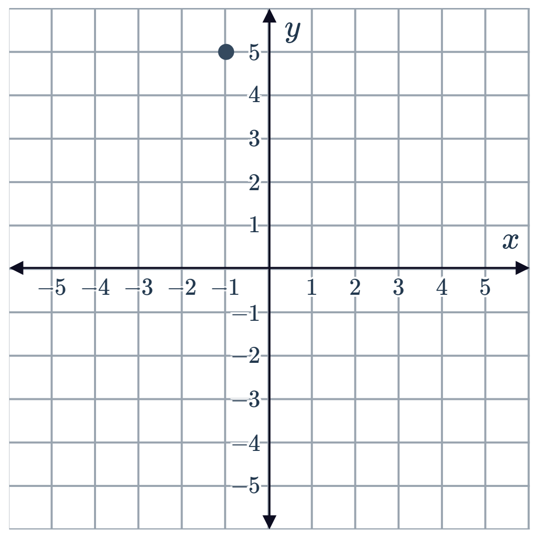
10 For the point on the plane, plot the point that will result from a translation of 5 units left followed by a rotation of 90° clockwise about the origin.



11 For the point on the plane, plot the point that will result from a translation of 2 units downwards followed by a rotation of 90° anticlockwise about the origin.

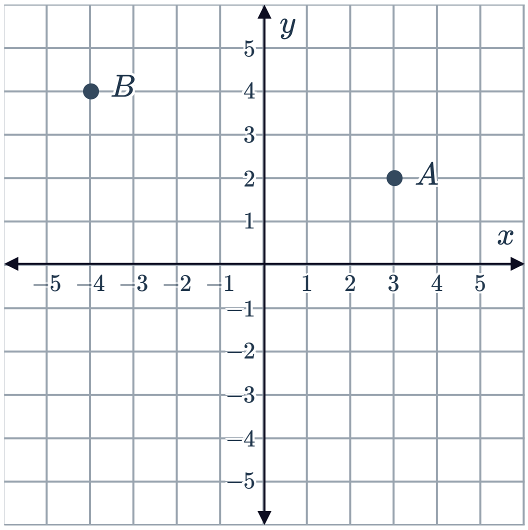


12 For the point on the plane, plot the point that will result from a translation of 3 units left and 4 units down, followed by a rotation of 90° anticlockwise about the origin.



13 Consider the two points, A and B on the plane:

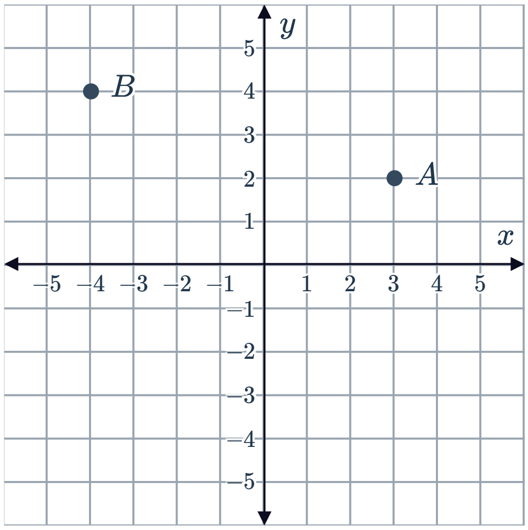
Plot the points that would result when A and B are first rotated 180° about the origin, and then reflected across the x-axis.



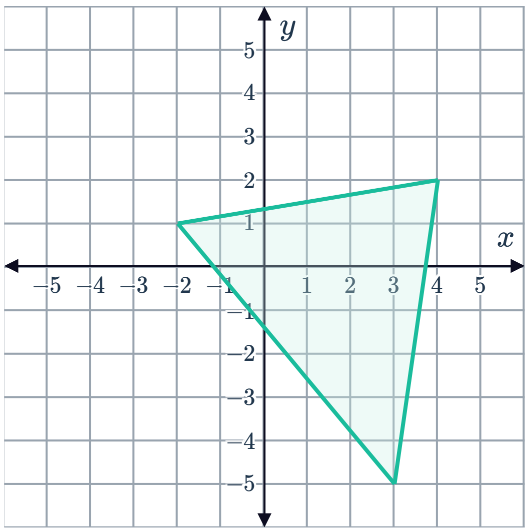
14 Consider the two points, A and B on the plane:

a Plot the points that would result when A and B are rotated 180° about the origin and reflected across the y-axis.

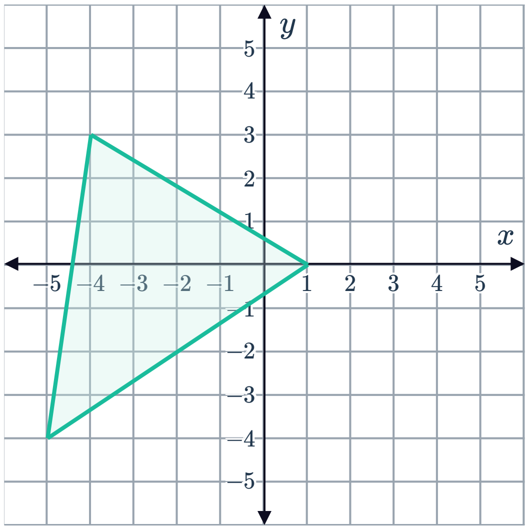
b What single transformation would give the same result?



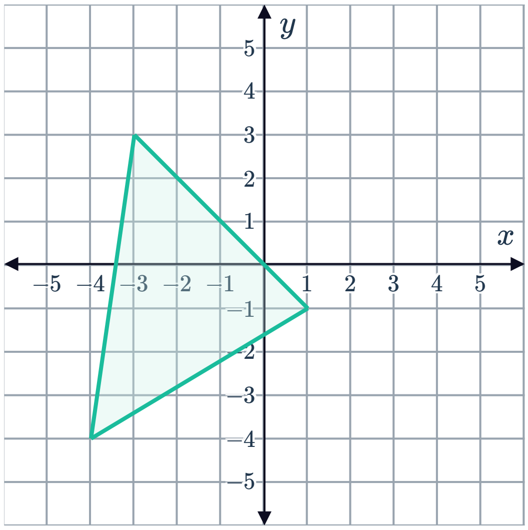
15 Plot the triangle that will result when the given triangle is reflected across the x-axis and translated 3 units to the left.



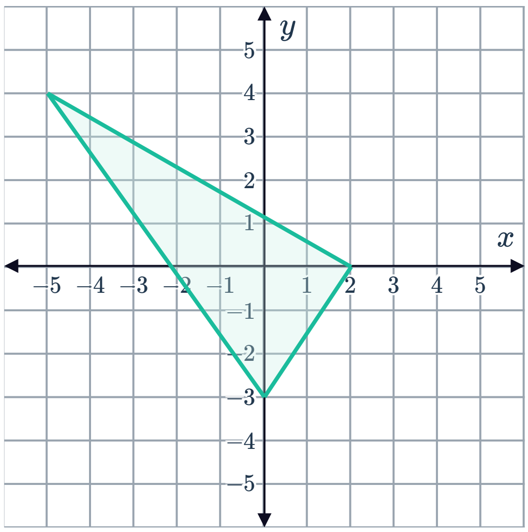
16 Plot the triangle that will result when the given triangle is reflected across the y-axis and translated 2 units up.



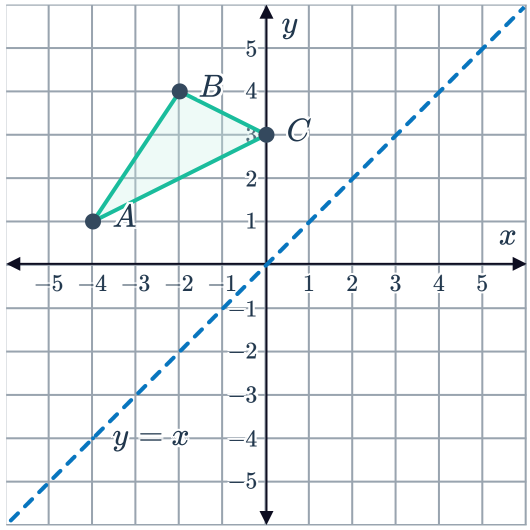
17 Plot the triangle that will result when the given triangle is reflected across the y-axis, and translated 3 units left and 2 units down.



18 Plot the triangle that will result when the given triangle is reflected across the x-axis, and translated 3 unit right and 2 units up.



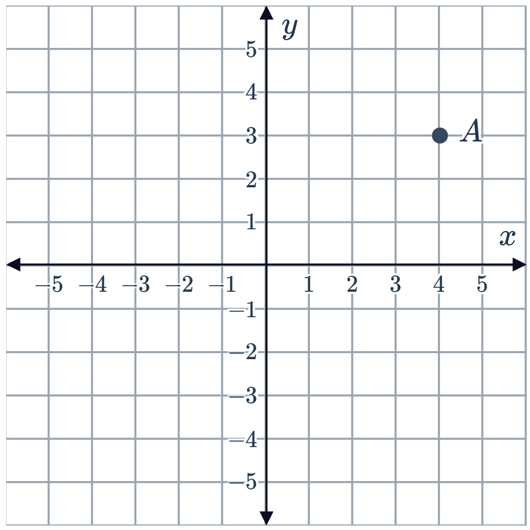
19 Consider the triangle ABC:



a Plot triangle $A'B'C'$, the result of reflecting triangle ABC across the line $y = x$.

b Suppose triangle $A'B'C'$ is reflected across the y-axis to form triangle $A''B''C''$. What single transformation would transform triangle ABC straight to triangle $A''B''C''$?

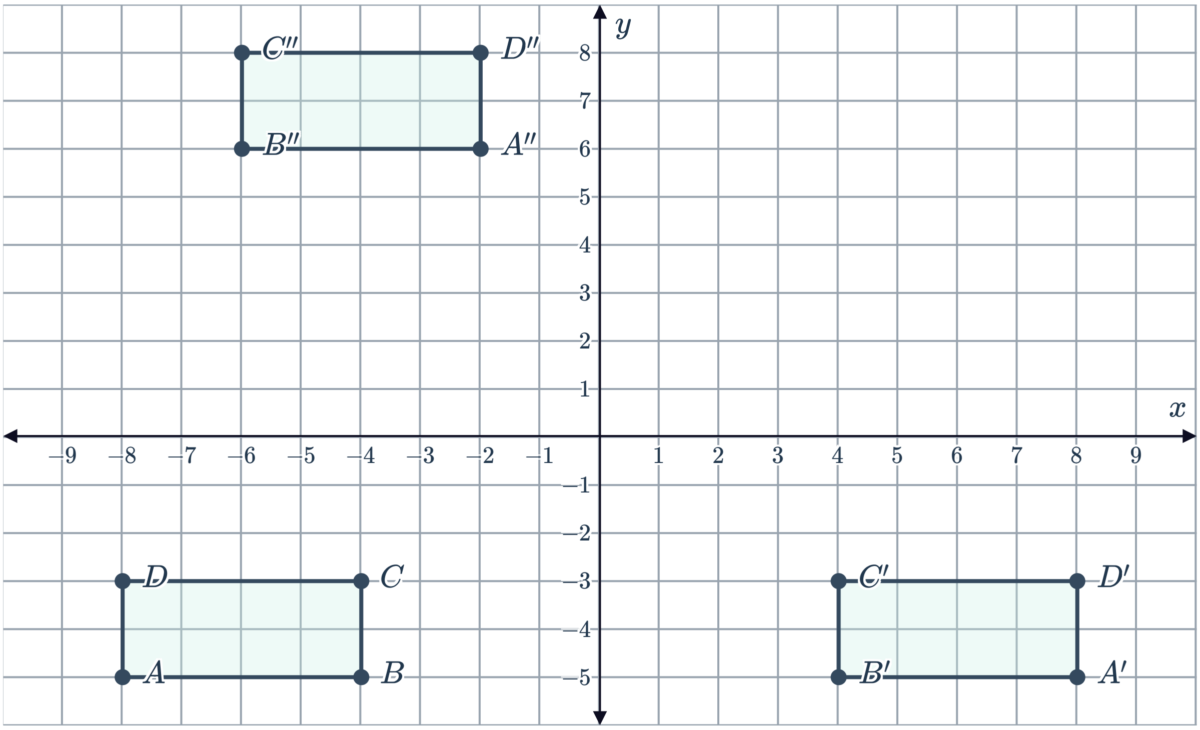
20 Point A is marked on the number plane:



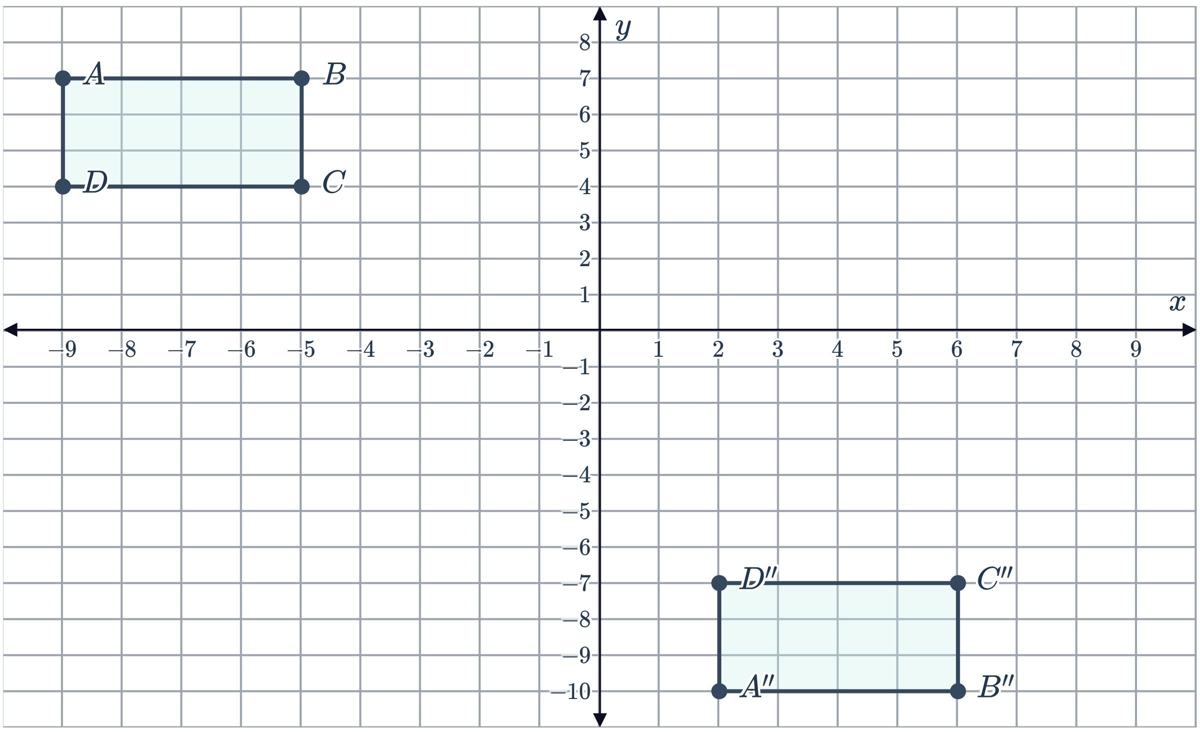
a If point A is translated 4 units down to make point B, what are the coordinates of point B?

b Across what line could point A be reflected across so that it makes point B?

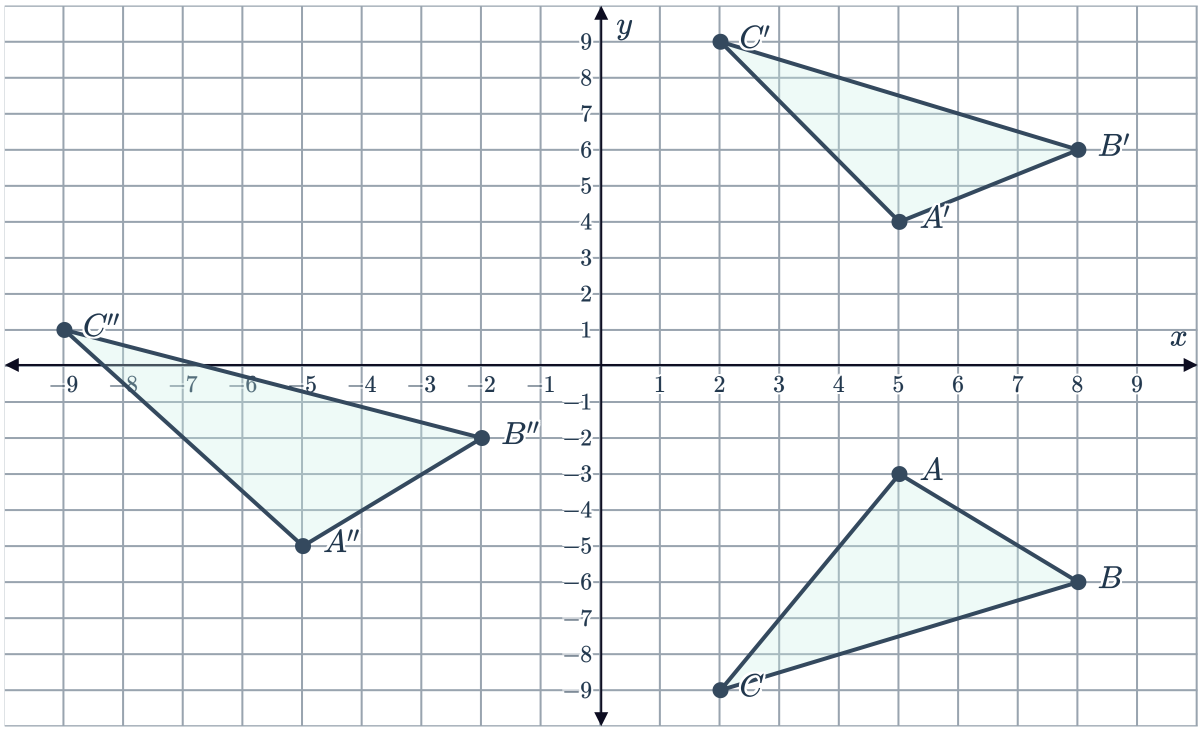
21 Describe the sequence of transformations [redacted]red to get from quadrilateral $ABCD$ to quadrilateral $A''B''C''D''$. Use quadrilateral $A'B'C'D'$ as a guide.



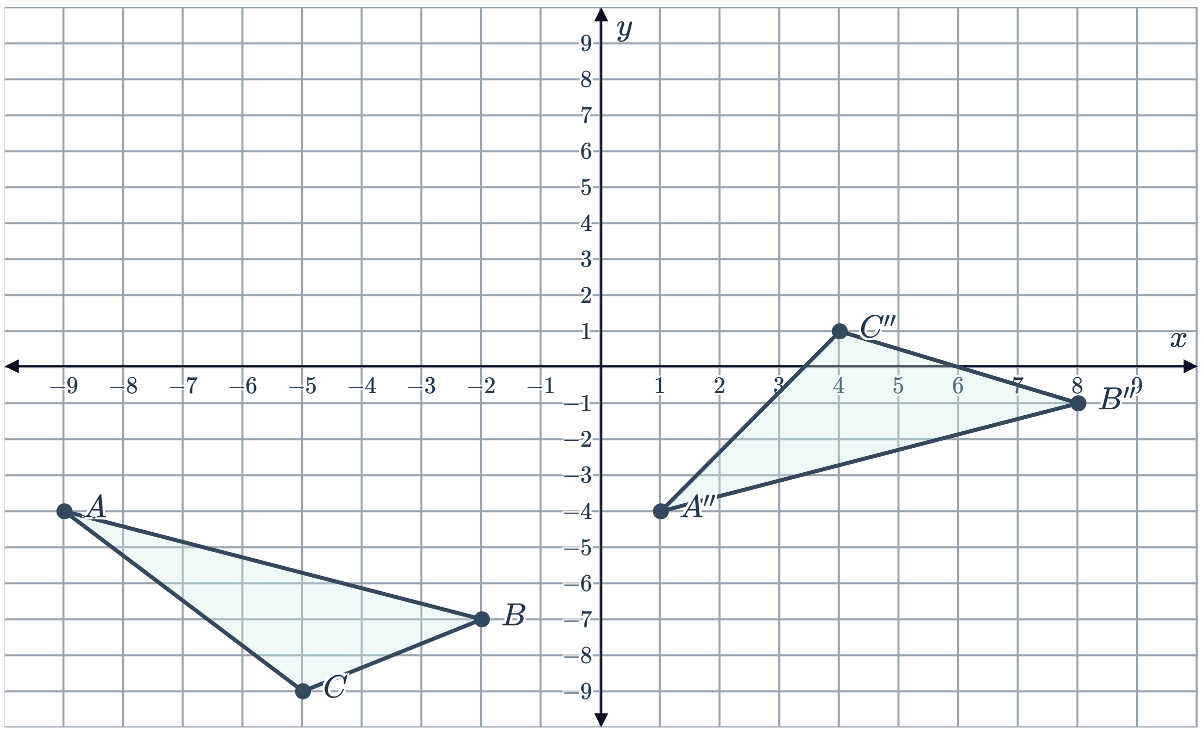
22 Describe the sequence of transformations required to get from quadrilateral $ABCD$ to quadrilateral $A''B''C''D''$.



23 Describe the sequence of transformations required to get from triangle ABC to triangle $A''B''C''$. Use triangle $A'B'C'$ as a guide.



24 Describe the sequence of transformations required to get from triangle ABC to triangle $A''B''C''$.



25 Point $A\,(-10, 6)$ is rotated 90° clockwise about the origin to form point B.

a Write down the the coordinates of point B.

b Point B is then translated to the left so that it lies on the same vertical line as point A. How many units to the left is point B translated?

26 For the point on the plane $(-6, 9)$, write d[redacted]he coordinates of the point that will result from a translation of 4 units left followed by a rotation of 90° clockwise about the origin.

27 For the point on the plane $(-2, -1)$, write down the coordinates of the point that will result from a translation of 2 units up followed by a rotation of 90° clockwise about the origin.

28 For each of the following points, write down the coordinates of the point that will result from a translation of 2 units right followed by a rotation of 90° anticlockwise about the origin:

a $(4, 5)$

b $(-7, -6)$

29 For the point on the plane $(1, 4)$, write down the coordinates of the point that will result from a translation of 4 units left and 3 units down, followed by a rotation of 90° clockwise about the origin.

30 For the point on the plane $(3, -6)$, write down the coordinates of the point that will result from a translation of 4 units right and 5 units down, followed by a rotation of 90° anticlockwise about the origin.

31 The point $A\,(6, -1)$is first rotated 180° about the origin, and then it is reflected across the x-axis to make point A'.

a Write down the coordinates ofA'.

b Describe a single transformation on$A$ that will produce the same result as part (a).

32 Consider the points: $A\,(-4, -4)$ and $B\,(8, 4)$

a Write down the coordinates of the points that would result when the original points are rotated 180° about the origin and reflected across the x-axis.

b Describe a single transformation on A and B that will produce the same result as part (a).

33 Consider the points: $A\,(-1, 2)$ and $B\,(-7, -3)$

a Write down the coordinates of the points that would result when the original points are rotated 180° about the origin and reflected across the y-axis.

b Describe a single transformation on A and B that will produce the same result as part (a).

34 Points $A\,(-4, -2)$, $B\,(-1, -9)$ and $C\,(-10, -6)$ are the vertices of a triangle. Write down the coordinates A', B' and C' that result from reflecting the triangle across the x-axis and translating it 4 units left.

35 Points $A\,(5, -9)$, $B\,(-1, -4)$ and $C\,(2, 5)$ are the vertices of a triangle. Write down the translating it 5 units down.

36 Points $A\,(2,-8)$, $B\,(8,5)$ and $C\,(7,6)$ are the vertices of a triangle. Write down the coordinates A', B' and C' that result from reflecting the triangle across the y-axis and translating it 4 units right.

37 Points $A\,(-9,-1)$, $B\,(10,5)$ and $C\,(10,1)$ are the vertices of a triangle. Write down the coordinates A', B' and C' that result from reflecting the triangle across the y-axis and translating it 4 units down.

38 Points $A\,(6,7)$, $B\,(-3,8)$ and $C\,(-3,10)$ are the vertices of a triangle. Write down the coordinates A', B' and C' that result from reflecting the triangle across the x-axis, and translating it 5 units left and 4 units down.

39 Points $A\,(4,-4)$, $B\,(10,3)$ and $C\,(9,9)$ are the vertices of a triangle. Write down the coordinates A', B' and C' that result from reflecting the triangle across the y-axis, and translating it 3 units right and 5 units down.

40 For any point (x,y) on the plane, write down the coordinates of the point after a rotation of 180° anticlockwise about the origin.

41 What two reflections result in the same transformation as rotating a point 180° anticlockwise about the origin?

42 For any point (a,b) on the plane, state the coordinates of the point after a reflection across the x-axis.

43 Consider the point $A\,(-9,0)$.

a If point A is rotated 90° clockwise about the origin to produce point B, find the coordinates of point B.

b Describe the two translations that can be used to move point A to point B.

44 Consider the point $A\,(-5,6)$.

a If point A is reflected across the y-axis to produce point B, and point B is then reflected across the x-axis to produce point C, find the coordinates of point C.

b Describe a single transformation to move point A to point C.